

IoT-Based Air Quality & Pollution Monitoring Dashboard

Overview

An industry-oriented IoT project that monitors environmental conditions in real time using ESP32, DHT22, MQ2, ThingSpeak Cloud, and Streamlit Dashboard.

The system collects air quality and environmental data, processes it, uploads it to the cloud, and visualizes insights through interactive dashboards.

Features

- Real-Time AQI Monitoring
 - Gas Leakage Detection
 - Temperature Monitoring
 - Humidity Monitoring
 - OLED Display Output
 - ThingSpeak Cloud Integration
 - Streamlit Analytics Dashboard
 - Pollution Alert Generation
 - CSV Data Logging
 - Historical Trend Analysis
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Technology Stack

Hardware

- ESP32
- MQ2 Gas Sensor
- DHT22 Temperature & Humidity Sensor
- OLED Display
- LEDs
- Buzzer

Software

- Arduino IDE
 - Python
 - Pandas
 - Streamlit
 - ThingSpeak
 - Wokwi Simulation
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Project Workflow

Sensors → ESP32 → AQI Processing → ThingSpeak Cloud → Streamlit Dashboard → Alerts & Reports

Key Features

AQI Classification

AQI Range	Status
0–100	Good
101–200	Moderate
201–300	Poor
301–500	Hazardous

Dashboard Metrics

- AQI
 - MQ2 Gas Level
 - Temperature
 - Humidity
 - Pollution Status
 - Alert Status
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Applications

- Smart Cities
 - Pollution Monitoring
 - Smart Homes
 - Hospitals
 - Schools and Colleges
 - Industrial Safety Monitoring
 - Environmental Research
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Learning Outcomes

- Embedded Systems Programming
 - ESP32 Development
 - IoT Communication Protocols
 - Cloud Integration
 - Dashboard Development
 - Data Analytics
 - Environmental Monitoring Systems
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Future Enhancements

- Email Alerts
 - SMS Notifications
 - AI-Based AQI Prediction
 - MQTT Integration
 - Node-RED Dashboard
 - Mobile Application Support
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Author

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Passionate about IoT, AI, Embedded Systems, Data Analytics, and Industry-Oriented Engineering Solutions.